

Validity Analysis: AFRIDOC MT

Target context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

Overall risk: **HIGH**

Deployment Context

Use case: Translating clinical practice guidelines, vaccination schedules, and maternal-health booklets into Amharic for ministry-of-health distribution

Target population: Health workers and patients across Ethiopian regional health bureaus producing and consuming Amharic-language material

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology	3	Moderate gaps	high
Input Content $\triangle$	2	Significant gaps	high
Input Form $\checkmark$	4	Minor gaps	high
Output Ontology $\triangle$	2	Significant gaps	high
Output Content $\triangle$	2	Significant gaps	high
Output Form $\checkmark$	4	Minor gaps	high
Average	2.8		

$\triangle$ = highest concern $\checkmark$ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

AFRIDOC-MT is a thoughtfully constructed ground-up document-level MT benchmark covering Amharic, with strong form match to the Ethiopia MOH deployment (text-only Amharic in Ethiopic script, document-level structure preserved, chrF prioritized for morphological richness, multi-parallel evaluation). However, three of four high-priority dimensions for this deployment have material gaps. Input Content (HIGH priority): the benchmark's WHO-sourced corpus contains substantial geographically remote and Ethiopia-irrelevant content (Afghanistan polio, CCHF, volcanoes, TRIPS), no Ethiopia-specific endemic-disease, calendar, or referral-pathway material, and visible translationese patterns. Output Ontology (HIGH priority): the deployment's primary success criterion — MOH/EFDA glossary compliance — is not operationalized by any benchmark

scoring function, and document-level term-consistency is only partially captured. Output Content (HIGH priority): the four named Amharic translators have no documented MOH credentials, clinical backgrounds, or regional affiliations; Krippendorff's $\alpha=0.46$ for Amharic is the lowest of the five languages; GPT-4o-as-judge is explicitly unreliable for Amharic. The maternal-health and immunization subset of the corpus does provide direct deployment-relevant content [DATASET-D4, D42, D44], and the benchmark's honest documentation of its own limitations (translationese, GPT-4o judge inconsistency, missing document-level QE for Amharic) is itself a significant strength.

ID	Evidence
D4	ተጨማሪ ምግብ ማጠቃለያ፡- በ 6 ወር አካባቢ የሕፃን የጋይል ፍላጎት ... ጨቅላ ህፃናት ... ጡት ማጥባት — Amharic complementary feeding article — maternal-child health topic directly relevant to deployment (maternal-health booklets)
D42	የዓለም ጤና ድርጅት ሕፃናትን ለ6 ወራት ብቻ እንዲያጠቡ ይመክራል ... ደህንነቱ የተጠበቀ እና ተጨማሪ ምግብ — Amharic WHO recommendation on exclusive breastfeeding — direct match for maternal-health booklet content
D44	ጨቅላ ህፃናት ከ6 ወር ጀምሮ የተጣራ፣ የተፈጨ እና ከፊል ጠጣር ምግቦችን መመገብ ይችላሉ — Amharic instruction for pureed/mashed foods from 6 months — clinically precise maternal-health instruction

Practical Guidance

What This Benchmark Measures

AFRIDOC-MT can validly measure (a) general document-level English→Amharic MT fluency and faithfulness on WHO-style health prose using d-chrF/d-BLEU [Q51, Q53], (b) relative model ranking among encoder-decoder NMT and decoder-only LLMs for Amharic [Q4, Q58], and (c) sentence-level vs. pseudo-document tradeoffs revealing that pseudo-document chunking can degrade African-language quality [Q61, Q72]. The benchmark provides a credible signal for whether a system can produce reasonably fluent, faithful Amharic translation of WHO-register health prose at document level.

Construct Depth

Construct depth is shallow for the deployment's primary criteria. Faithfulness and surface fluency are reasonably well covered through d-chrF, human DA, and CE/fluency dimensions [Q183, Q195]. Document-level discourse phenomena are partially probed via lexical and grammatical cohesion error categories [Q189, Q190], though GPT-4o reliability for these is poor in Amharic [Q99, Q219]. The benchmark does NOT probe institutional terminology compliance, clinical-accuracy validation by MOH-credentialed reviewers, regional Amharic register acceptance, or readability for semi-literate patients — all of which are central to deployment success.

What Else You Need

Five concrete supplements are needed: (1) MOH/EFDA glossary-conformance scoring against the Academy of Ethiopian Languages Amharic Medical Dictionary [WEB-8] and any EFDA list — addresses OO gap; (2) parallel reference translations or post-editing by MOH-credentialed clinical translators on a held-out Ethiopia MOH-distributed sample — addresses OC gap; (3) Ethiopia-specific extension corpus covering endemic diseases (malaria, TB, kala-azar, trachoma), Ethiopian calendar dates, and tiered-referral language — addresses IC gap; (4) regional bureau acceptance review with stakeholders from Tigray, Oromia, Somali, Amhara bureaus on register acceptability — addresses OC regional-register gap; (5) readability evaluation with a sample of semi-literate patients or HEW-mediated comprehension testing — addresses OO readability calibration gap.

ID	Evidence
Q4	"Our results indicate that NLLB-200 achieves the best average performance among the standard NMT models, while GPT-4o outperforms general-purpose LLMs." (p.1)
Q51	"Hence, we realigned sentence-level or pseudo-translation outputs into full documents, then computed BLEU (Papineni et al., 2002) and chrF (Popović, 2015) to create document BLEU (d-BLEU) and document chrF (d-chrF)." (p.5)
Q53	"We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C." (p.5)
Q58	"On average the NLLB models obtain scores of 65.4/66.6 and 64.3/65.0 on health and tech domains respectively, with 3.3B outperforming 1.3B except when translating into Yorùbá." (p.5)

Q61	"Pseudo-document translation is worse than sentence-level translation when translating into African languages Our results from pseudo-document translation show a performance drop across different models compared to sentence-level translation, especially when translating into African languages." (p.6)
Q72	"Human evaluation results align closely with d-chrF, which favors sentence-level translations over pseudo-document translations when translating into African languages." (p.8)
Q99	"GPT-4o was employed to assess and rate the translation outputs of four models. While its ratings were consistent for translations into English, the same was not observed for translations into African languages, likely due to the model's limited understanding of these languages." (p.10)
Q183	"GPT-4 is prompted to identify and list the mistakes, such as incorrect translations, omissions, additions, and any other errors, by comparing the model's output to the reference translation." (p.22)
Q189	"Lexical Cohesion Mistakes: Issues with vocabulary usage, incorrect or missing synonyms, or overuse of certain words that disrupt the flow." (p.23)
Q190	"Grammatical Cohesion Mistakes: Problems with pronouns, conjunctions, or grammatical structures that link sentences and clauses." (p.23)
Q195	"Each annotator was assigned 80 documents to evaluate, tasked with marking as many error spans as possible and rating the overall quality on a scale from 0 to 100." (p.24)
Q219	"These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages." (p.26)
WEB-8	Academy of Ethiopian Languages Amharic Medical Dictionary exists as a potentially relevant terminology reference for re-annotation (https://archive.org/details/amharic-dictionary-of-medical-terms-108p-copy)

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: AFRIDOC_MT | Context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

AFRIDOC-MT is a ground-up document-level MT benchmark designed for African languages including Amharic, with a health domain category that aligns with the deployment's prose-narrative scope [Q1, Q2, Q15]. The user has explicitly scoped structured document types (tables, timetables, forms) as out-of-scope, which mitigates the most obvious category gap. However, the benchmark mixes a tech-news category (Techpoint Africa) [Q18, Q22] that introduces construct-irrelevant variance for clinical deployments, and within the health category a substantial portion of sampled documents covers topics geographically or substantively distant from Ethiopian MOH clinical priorities (Afghanistan polio, CCHF, volcanoes, Zika, TRIPS policy) [DATASET-D7, D32, D33, D34, D26, D46]. Multi-way translation across 30 directions is supported but the deployment uses only en→am [Q7, Q94, Q101]. Net: the taxonomy partially aligns — document-level MT for Amharic prose health is covered, but the benchmark's category structure does not isolate clinical-guideline / maternal-health / vaccination subgenres that the deployment actually translates.

ID	Evidence
Q1	"This paper introduces AFRIDOC-MT, a document-level multi-parallel translation dataset covering English and five African languages: Amharic, Hausa, Swahili, Yorùbá, and Zulu." (p.1)
Q2	"The dataset comprises 334 health and 271 information technology news documents, all human-translated from English to these languages." (p.1)
Q7	"In addition, AFRIDOC-MT supports multi-way translation, allowing translations not only between English and the African languages but also between any two of the languages covered." (p.1)
Q15	"Document-level NMT aims to overcome the limitations of sentence-level systems by translating an entire document as a whole." (p.2)
Q18	"We scraped English articles from the websites of Techpoint Africa and the World Health Organization (WHO). The articles cover different topics of different lengths with an average length of 30 and 37 sentences for health and tech respectively." (p.3)
Q22	"Finally, we selected 334 and 271 English articles/documents from the health and tech domains respectively, which represents 10k sentences each per domain." (p.3)
Q94	"AFRIDOC-MT is a multi-way parallel dataset. However, due to the cost of running inference over three prompts and across all 30 translation directions for all the models evaluated, most of our analysis is limited to translation tasks between English and the five African languages." (p.10)
Q101	"While we fine-tuned both NLLB-1.3B and LLaMAX3 models across all 30 language directions, due to computational constraints and the high cost of qualitative evaluation, our detailed analysis focuses only on translation between English and the 5 African languages." (p.10)
D7	ነገር ግን፣ በ2023 እስከ ዛሬ 46 ደብሊዊፒቪ1 አዎንታዊ የአካባቢ ናሙናዎች ... ናንጋርሃር እና ኩናር አውራጃዎች ... ሚሊዮን ተመላሾች — Amharic WHO article on Afghanistan polio — geographic specificity to Afghanistan/Pakistan, not Ethiopia; pseudo-document chunk (10 sentences)
D26	ዚካ ቫይረስ ... ፀረ-ተባይ ... Vector control operations framework for Zika virus — Short Zika virus sentence fragment — topic not endemic to Ethiopia
D32	Crimean-Congo haemorrhagic fever (CCHF) is a viral haemorrhagic fever usually transmitted by ticks ... CCHF is endemic in all of Africa, the Balkans, the Middle East and in Asia. — CCHF article — hemorrhagic fever not a primary health priority in Ethiopia; tick-borne virus
D33	A volcano is a vent in the Earth's crust from which eruptions occur... There are about 1500 potentially active volcanoes worldwide. — Volcanic eruption WHO article — not relevant to Ethiopian clinical translation deployment

D34	Afghanistan continues to implement an intense campaign schedule, focusing on improving quality in the endemic zone... massive population movement significantly increases the risk of cross-border poliovirus spread — WHO Afghanistan polio article — geographically specific to Afghanistan, not Ethiopia
D46	TRIPS Agreement, many resolutions of the World Health Assemblies have requested WHO to address the impact of trade agreements and intellectual property protection on public health and access to medicines — TRIPS/IP health policy — highly technical policy content

Strengths

- Ground-up design for African languages including Amharic with document-level MT as a distinct construct from sentence-level MT [Q1, Q15, Q16]
- Health domain category is present and operationalized at document level — the closest existing category match for the deployment [Q2, Q14, DATASET-D4, D15, D42]
- Deployment scope (prose health) aligns with the benchmark's narrative-prose-only corpus structure [Q18, DATASET-D1, D3]

ID	Evidence
Q1	"This paper introduces AFRIDOC-MT, a document-level multi-parallel translation dataset covering English and five African languages: Amharic, Hausa, Swahili, Yorùbá, and Zulu." (p.1)
Q2	"The dataset comprises 334 health and 271 information technology news documents, all human-translated from English to these languages." (p.1)
Q14	"To the best of our knowledge, only TICO-19, a health domain translation benchmark, has the potential to be used for document-level MT, while it is restricted to topics related to COVID-19." (p.2)
Q15	"Document-level NMT aims to overcome the limitations of sentence-level systems by translating an entire document as a whole." (p.2)
Q16	"Both document-level and context-aware MT allow for the possibility of improving translation quality for context-dependent phenomena such as coreference resolution (Müller et al., 2018; Bawden et al., 2018; Voita et al., 2018; Herold and Ney, 2023), lexical disambiguation (Rios Gonzales et al., 2017; Martínez Garcia et al., 2019), and lexical cohesion (Wong and Kit, 2012; Garcia et al., 2014, 2017; Bawden et al., 2018; Voita et al., 2019)." (p.2)
Q18	"We scraped English articles from the websites of Techpoint Africa and the World Health Organization (WHO). The articles cover different topics of different lengths with an average length of 30 and 37 sentences for health and tech respectively." (p.3)
D1	የሞቃታማ ዐውሎ ነፋሶች ... ለሞቃታማ አውሎ ነፋሶች፣ እንዲሁም ሀይለኛ አውሎ ነፋሶ ወይም አውሎ ነፋሶች ... ኃይለኛ ክብ አውሎ ነፋሶች ... 233 000 ሰዎችን ገድለዋል — Full Amharic document on tropical cyclones (WHO article), multi-sentence, Ethiopic script, with clinical/epidemiological content and specific numbers
D3	አካል ጉዳተኝነት ሰው የመሆን አካል ነው። ሁሉም ሰው ማለት ይቻላል በሕይወታቸው ላይ በሆነ ወቅት ጊዜያዊ ወይም ዘላቂ የአካል ጉዳት ያጋጥማቸዋል — Amharic WHO article on disability — prose narrative, full document preserved
D4	ተጨማሪ ምግብ ማጠቃለያ፡- በ 6 ወር አካባቢ የሕፃን የኃይል ፍላጎት ... ጨቅላ ህፃናት ... ጡት ማጥባት — Amharic complementary feeding article — maternal-child health topic directly relevant to deployment (maternal-health booklets)
D15	ደህንነቱ የተጠበቀ፡- በንጽህና ተከማችተው እና ተዘጋጅተዋል እንዲሁም ጠርመሶች እና ፕላስቲኮችን ሳይሆን በንጹህ እቃዎች ... ጡት ማጥባት — Complementary feeding clinical guidance in Amharic — high relevance (maternal health booklet content)
D42	የዓለም ጤና ድርጅት ሕፃናትን ለ6 ወራት ብቻ እንዲያጠቡ ይመክራል ... ደህንነቱ የተጠበቀ እና ተጨማሪ ምግብ — Amharic WHO recommendation on exclusive breastfeeding — direct match for maternal-health booklet content

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for this deployment are clinical practice guidelines (narrative), maternal-health booklets, and vaccination schedule narratives — all prose. Benchmark provides a 'health' category from WHO articles that partially covers these subgenres [Q2, Q18, DATASET-D4, D9, D42].

IO-2: The benchmark does not subdivide health into clinical-guideline / maternal-health / immunization-schedule / endemic-disease subcategories. Ethiopia-specific subgenres (e.g., MOH referral pathways, endemic-disease guidance) are absent — only TICO-19 is mentioned as a comparable health document-level resource and is COVID-restricted [Q14]. The deployment-relevant maternal-child / immunization subset is present but constitutes a minority of sampled health documents [DATASET-D4, D15, D42, D31, D44].

IO-3: The benchmark includes a tech-news category (Techpoint Africa, 271 documents) that is irrelevant to clinical translation deployment [Q2, Q18, DATASET-D13, D14, D22, D38, D39]. Within health, a substantial share of sampled WHO documents covers globally generic or non-Ethiopian-relevant topics (Afghanistan polio, CCHF, volcanoes, TRIPS, IHR, Zika) [DATASET-D7, D32, D33, D34, D46, D26].

IO-4: Two category-level issues harm content validity: (a) inclusion of irrelevant tech-domain category that dilutes any aggregate score [DATASET-D14, D22], and (b) absence of fine-grained clinical-guideline / maternal-health / endemic-disease subcategorization within the health domain, meaning a system can score well on aggregate health while underperforming on the deployment-specific subgenres [DATASET-D7, D33].

ID	Evidence
Q2	"The dataset comprises 334 health and 271 information technology news documents, all human-translated from English to these languages." (p.1)
Q14	"To the best of our knowledge, only TICO-19, a health domain translation benchmark, has the potential to be used for document-level MT, while it is restricted to topics related to COVID-19." (p.2)
Q18	"We scraped English articles from the websites of Techpoint Africa and the World Health Organization (WHO). The articles cover different topics of different lengths with an average length of 30 and 37 sentences for health and tech respectively." (p.3)
D4	ተጨማሪ ምግብ ማጠቃለያ፡- በ 6 ወር አካባቢ የሕፃኑ የኃይል ፍላጎት ... ጨቅላ ህፃናት ... ጡት ማጥጣት — Amharic complementary feeding article — maternal-child health topic directly relevant to deployment (maternal-health booklets)
D7	ነገር ግን፡ በ2023 እስከ ዛሬ 46 ደብሊዊፒቪ1 አዎንታዊ የአካባቢ ናሙናዎች ... ናንጋርሃር እና ኩናር አውራጃዎች ... ሚሊዮን ተመላሾች — Amharic WHO article on Afghanistan polio — geographic specificity to Afghanistan/Pakistan, not Ethiopia; pseudo-document chunk (10 sentences)
D9	ዓለም አቀፍ ትብብር የጋራ ዓላማን ለማሳካት ... ዜጅ ዶዝ ... ህፃናት ፈልጎ ማግኘት ... ዶ/ር ጆርጅ ምዊንያ — Amharic WHO Assembly article about immunization — broad vaccination language relevant to vaccination schedule booklets
D13	ፔይ ዲይ ለሽያጭ ? ዛሬ በ ቴክ ፖይንት አፍሪካ ... የየይጂሪያ ፕረዚዳንት ... ፔይ ዲይ ... ኦባሲ እነ-አቦንግ — Amharic Techpoint Africa tech-news article about Nigerian fintech/genomics — not relevant to Ethiopian MOH deployment
D14	ደቡብ አፍሪካ ጎጂ ይዘቶችን በመስመር ላይ ማቆም ትፈልጋለች ... YouTube ... NALA ... Cellulant ... ሴሉላንት 20% — Long Amharic tech article: South Africa content regulation, YouTube, NALA remittances, Cellulant layoffs — entirely irrelevant to Ethiopian clinical translation
D15	ደህንነቱ የተጠበቀ፡- በንጽህና ተከማችተው እና ተዘጋጅተዋል እንዲሁም ጠርመሶች እና ፕላስቲኮችን ሳይሆን በንጹህ እቃዎች ... ጡት ማጥጣት — Complementary feeding clinical guidance in Amharic — high relevance (maternal health booklet content)
D22	Fatanmi identifies its effect on marketing a business ... Gen Zs ... Building in public... — Tech article about startup marketing strategy — entirely irrelevant to Ethiopian MOH clinical translation deployment
D26	ዚካ ቫይረስ ... ፀረ-ተባይ ... Vector control operations framework for Zika virus — Short Zika virus sentence fragment — topic not endemic to Ethiopia
D31	ጡት ማጥጣት ... ሞተዋል ... 820,000 ... 40% ... — Amharic infant nutrition WHO article — directly relevant to maternal-health booklet
D32	Crimean-Congo haemorrhagic fever (CCHF) is a viral haemorrhagic fever usually transmitted by ticks ... CCHF is endemic in all of Africa, the Balkans, the Middle East and in Asia. — CCHF article — hemorrhagic fever not a primary health priority in Ethiopia; tick-borne virus
D33	A volcano is a vent in the Earth's crust from which eruptions occur... There are about 1500 potentially active volcanoes worldwide. — Volcanic eruption WHO article — not relevant to Ethiopian clinical translation deployment
D34	Afghanistan continues to implement an intense campaign schedule, focusing on improving quality in the endemic zone... massive population movement significantly increases the risk of cross-border poliovirus spread — WHO Afghanistan polio article — geographically specific to Afghanistan, not Ethiopia
D38	የሕንድ ኢድቴክ ኢንዱስትሪ ... ሕንድ ከዩናይትድ ስቴትስ (ዩ ኤስ) ቀጥሎ ... edtech ... Central Square Foundation ... Nigerian Edtech ecosystem — Indian/Nigerian edtech article — entirely irrelevant to Ethiopian clinical translation

D39	Nigerian healthtech startup, Clafiya, raises \$610,000 in pre-seed funding — African healthtech/fintech startup news — entirely irrelevant to Ethiopian MOH clinical translation deployment
D42	የዓለም ጤና ድርጅት ሕፃናትን ለ6 ወራት ብቻ እንዲያጠቡ ይመክራል ... ደህንነቱ የተጠበቀ እና ተጨማሪ ምግብ — Amharic WHO recommendation on exclusive breastfeeding — direct match for maternal-health booklet content
D44	ጨቅላ ህፃናት ከ6 ወር ጀምሮ የተጣራ፣ የተፈጨ እና ከፊል ጠጣር ምግቦችን መመገብ ይችላሉ — Amharic instruction for pureed/mashed foods from 6 months — clinically precise maternal-health instruction
D46	TRIPS Agreement, many resolutions of the World Health Assemblies have requested WHO to address the impact of trade agreements and intellectual property protection on public health and access to medicines — TRIPS/IP health policy — highly technical policy content
WEB-10	JMIR 2025 scoping review identified only 3 Amharic NLP-for-public-health studies; none evaluated Ethiopia-endemic disease terminology (https://pmc.ncbi.nlm.nih.gov/articles/PMC11923465/)

Information Gaps

- Proportion of WHO health documents that are clinical-guideline / maternal-health / immunization vs. policy / financing / non-Ethiopian-endemic — not quantified beyond 51-document sample
- Whether vaccination schedule narrative (a deployment subgenre) is meaningfully present in the 334-document health corpus

Requires Expert Verification

- Whether the AFRIDOC-MT health subgenre distribution adequately represents the document types Ethiopia MOH actually distributes (requires comparison with MOH document inventory)

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: AFRIDOC_MT | Context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's source content is generic English WHO articles, not Ethiopia-specific clinical material. The user accepts generic coverage as sufficient for current scope, but the deployment serves a context with Ethiopia-specific endemic conditions (kala-azar, trachoma, malaria), the Ethiopian calendar, and tiered referral pathways — all explicitly absent [Q103, Q105]. Dataset analysis confirms a substantial share of sampled health documents cover topics geographically/contextually remote from Ethiopia [DATASET-D7, D32, D33, D34, D26]. The benchmark itself flags translationese risk — translated Amharic sentences may exhibit unnatural syntax, overly literal phrasing, and English-centric structural and cultural framing [Q103, Q104, Q105], and dataset inspection confirms visible English-loanword and parenthetical-acronym patterns in Amharic [DATASET-D43, D47]. With IC marked HIGH priority by the user, these gaps drive a score of 2: significant construct-irrelevant variance from non-Ethiopian topics combined with Ethiopia-specific construct underrepresentation.

ID	Evidence
Q103	"A potential limitation of our dataset is the influence of translationese (Koppel and Ordan, 2011). Since all source material translated originates in English, translated sentences in African languages may exhibit patterns such as unnatural syntax or overly literal phrasing." (p.10)
Q104	"Although we have not conducted an analysis to quantify these effects, prior work suggests that they can affect MT model performance, generalization and evaluation including direct assessment (Freitag et al., 2019; Edunov et al., 2020)." (p.10)
Q105	"Furthermore, AFRIDOC-MT may reflect a bias toward English in terms of structure, semantics, and cultural framing. We leave a deeper investigation of these issues to future work." (p.10)
D7	ነገር ግን፣ በ2023 እስከ ዛሬ 46 ደብሊዊፒቪ1 አዎንታዊ የአካባቢ ናሙናዎች ... ናንጋርሃር እና ኩናር አውራጃዎች ... ሚሊዮን ተመለሾች — Amharic WHO article on Afghanistan polio — geographic specificity to Afghanistan/Pakistan, not Ethiopia; pseudo-document chunk (10 sentences)
D26	ዚካ ቫይረስ ... ፀረ-ተባይ ... Vector control operations framework for Zika virus — Short Zika virus sentence fragment — topic not endemic to Ethiopia
D32	Crimean-Congo haemorrhagic fever (CCHF) is a viral haemorrhagic fever usually transmitted by ticks ... CCHF is endemic in all of Africa, the Balkans, the Middle East and in Asia. — CCHF article — hemorrhagic fever not a primary health priority in Ethiopia; tick-borne virus
D33	A volcano is a vent in the Earth's crust from which eruptions occur... There are about 1500 potentially active volcanoes worldwide. — Volcanic eruption WHO article — not relevant to Ethiopian clinical translation deployment
D34	Afghanistan continues to implement an intense campaign schedule, focusing on improving quality in the endemic zone... massive population movement significantly increases the risk of cross-border poliovirus spread — WHO Afghanistan polio article — geographically specific to Afghanistan, not Ethiopia
D43	ፀረ-ተህዋሲያን - ፀረ-ባክቴሪያ ፣ፀረ-ቫይረስ፣ ፀረ-ፈንገስ ... 'ሱፐርባግስ(superbugs)' — AMR article with English loanword "superbugs" retained in Amharic — illustrates translationese patterns
D47	CCHF ወረርሽኞች ቫይረሱ ወደ ወረርሽኞች ሊያመራ ስለሚችል ከፍተኛ የሞት መጠን (ከ10-40%)... ህክምና፡- የክራይሚያ-ኮንጎ ... ፀረ-ቫይረስ መድሃኒት ... ribavirin — CCHF treatment article in Amharic including "ribavirin" (transliterated drug name) — medical terminology rendering pattern visible

Strengths

- WHO-sourced register matches the institutional voice of MOH-distributed health communication materials [Q18, DATASET-D11, D41]

- Maternal-child health and immunization subset (complementary feeding, breastfeeding, vaccination) is directly aligned with deployment document types [DATASET-D4, D15, D42, D44]
- Authors document data-selection ethics with attention to bias minimization and harmful-content screening [Q106, Q107]

ID	Evidence
Q18	"We scraped English articles from the websites of Techpoint Africa and the World Health Organization (WHO). The articles cover different topics of different lengths with an average length of 30 and 37 sentences for health and tech respectively." (p.3)
Q106	"AFRIDOC-MT was created with the utmost consideration for ethical standards. The English texts translated were sourced from publicly available and ethically sourced materials. The data sources were selected to represent different cultural perspectives, with a focus on minimizing any potential bias." (p.10)
Q107	"Efforts were made to ensure the dataset does not include harmful, biased, or offensive content via manual inspection." (p.10)
D4	ተጨማሪ ምግብ ማጠቃለያ፡- በ 6 ወር አካባቢ የሕፃን የኃይል ፍላጎት ... ጨቅላ ህፃናት ... ጡት ማጥባት — Amharic complementary feeding article — maternal-child health topic directly relevant to deployment (maternal-health booklets)
D11	Measures for the prevention of cholera mostly consist of providing clean water and proper sanitation to populations who do not yet have access to basic services, as well as vaccination with Oral Cholera Vaccines. — Sentence-level WHO cholera prevention content — cholera is endemic in Ethiopia but article is generic global
D15	ደህንነቱ የተጠበቀ፡- በንጽህናና ተከማቻተው እና ተዘጋጅተዋል እንዲሁም ጠርቦሶች እና ፕላስቲኮችን ሳይሆን በንጹህ እቃዎች ... ጡት ማጥባት — Complementary feeding clinical guidance in Amharic — high relevance (maternal health booklet content)
D41	WHO helps to restore primary care services so that facilities can deliver essential services, including immunization, basic treatment for common illnesses, acute malnutrition and maternal care — WHO cyclone response includes immunization + maternal care — partially relevant content
D42	የዓለም ጤና ድርጅት ሕፃናትን ለ6 ወራት ብቻ እንዲያጠቡ ይመክራል ... ደህንነቱ የተጠበቀ እና ተጨማሪ ምግብ — Amharic WHO recommendation on exclusive breastfeeding — direct match for maternal-health booklet content
D44	ጨቅላ ህፃናት ከ6 ወር ጀምሮ የተጣራ፣ የተፈጠሰ እና ከፊል ጠጣር ምግቦችን መመገብ ይችላሉ — Amharic instruction for pureed/mashed foods from 6 months — clinically precise maternal-health instruction

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: The deployment requires Ethiopia-specific knowledge: endemic diseases (malaria, TB, kala-azar, trachoma), Ethiopian calendar dates, MOH referral pathways, regional health bureau structures. None of these is represented in the benchmark — sampled documents include Afghanistan polio, CCHF, Zika, and volcanoes, none Ethiopia-endemic [DATASET-D7, D32, D26, D33]. Web search confirmed no NLP study has evaluated Amharic MT on Ethiopia-endemic disease terminology [WEB-10].

IC-2: Source documents are publicly available WHO and Techpoint articles described as ethically sourced and screened for harmful content [Q34, Q106, Q107]. Cultural alignment with Ethiopia is partial: WHO health register is broadly compatible, but Ethiopian cultural context (religious diversity, household decision-making, Ethiopian Orthodox / Muslim practices around maternal health) is not specifically reflected.

IC-3: The benchmark explicitly acknowledges potential bias toward English in structure, semantics, and cultural framing [Q105]. Dataset inspection shows English-centric patterns: parenthetical English acronyms (AMR, CCHF), English loanwords ('superbugs'), and direct transliterations of drug names (ribavirin) embedded in Amharic [DATASET-D43, D5, D6, D47]. The tech-domain content is entirely Western/Nigerian/Indian and irrelevant [DATASET-D14, D22, D38].

IC-4: INSUFFICIENT DOCUMENTATION — the benchmark does not document recruitment of Ethiopian MOH clinical reviewers to vet content for Ethiopian relevance. Translators are named but no review of source-content cultural fit is described. Would need MOH expert review of the 334 health documents against Ethiopia MOH distribution material to confirm content alignment.

IC-5: Two main content-validity harms: (1) construct underrepresentation — Ethiopia-specific clinical content (endemic diseases, calendar, referral pathways) is absent [WEB-10]; (2) construct-irrelevant variance — non-Ethiopian-relevant topics (Afghanistan polio, volcanoes, CCHF) dilute the evaluation signal [DATASET-D7, D32, D33], and translationese effects may produce English-shaped Amharic that scores well against references but reads unnaturally [Q103, DATASET-D43].

ID	Evidence
Q34	"The health data is licensed under CC BY-NCSA 3.0, while the tech data is licensed under CC BY-NC-SA 4.0." (p.4)
Q103	"A potential limitation of our dataset is the influence of translationese (Koppel and Ordan, 2011). Since all source material translated originates in English, translated sentences in African languages may exhibit patterns such as unnatural syntax or overly literal phrasing." (p.10)
Q105	"Furthermore, AFRIDOC-MT may reflect a bias toward English in terms of structure, semantics, and cultural framing. We leave a deeper investigation of these issues to future work." (p.10)
Q106	"AFRIDOC-MT was created with the utmost consideration for ethical standards. The English texts translated were sourced from publicly available and ethically sourced materials. The data sources were selected to represent different cultural perspectives, with a focus on minimizing any potential bias." (p.10)
Q107	"Efforts were made to ensure the dataset does not include harmful, biased, or offensive content via manual inspection." (p.10)
D5	ፀረ-ተሕዋስያንን መቋቋም(AMR) ... ባክቴሪያ፣ ጥገኛ ተውሳኮች፣ ቫይረሶችና በፈንገሶች ምክንያት — Amharic AMR article — specialized medical terminology in Ethiopic script
D6	ክራይሚያ-ኮንጎ የደም መፍሰስ ትኩሳት ... CCHF ወረርሽኞች ... ከ10-40% ሞት ... ሞቃታማ አውሎ ነፋስ — Amharic article on CCHF hemorrhagic fever — clinical terminology, disease outside typical Ethiopian endemic profile
D7	ነገር ግን፣ በ2023 እስከ ዛሬ 46 ደብሊዊፒቪ1 አዎንታዊ የአካባቢ ናሙናዎች ... ናጋርሃር እና ኩናር አውራጃዎች ... ሚሊዮን ተመላሾች — Amharic WHO article on Afghanistan polio — geographic specificity to Afghanistan/Pakistan, not Ethiopia; pseudo-document chunk (10 sentences)
D14	ደቡብ አፍሪካ ጎጂ ይዘቶችን በመስመር ላይ ማቆም ትፈልጋለች ... YouTube ... NALA ... Cellulant ... ሴላጎንት 20% — Long Amharic tech article: South Africa content regulation, YouTube, NALA remittances, Cellulant layoffs — entirely irrelevant to Ethiopian clinical translation
D22	Fatanmi identifies its effect on marketing a business ... Gen Zs ... Building in public... — Tech article about startup marketing strategy — entirely irrelevant to Ethiopian MOH clinical translation deployment
D26	ዚካ ቫይረስ ... ፀረ-ተባይ ... Vector control operations framework for Zika virus — Short Zika virus sentence fragment — topic not endemic to Ethiopia
D32	Crimean-Congo haemorrhagic fever (CCHF) is a viral haemorrhagic fever usually transmitted by ticks ... CCHF is endemic in all of Africa, the Balkans, the Middle East and in Asia. — CCHF article — hemorrhagic fever not a primary health priority in Ethiopia; tick-borne virus
D33	A volcano is a vent in the Earth's crust from which eruptions occur... There are about 1500 potentially active volcanoes worldwide. — Volcanic eruption WHO article — not relevant to Ethiopian clinical translation deployment
D38	የሕንድ ኢ-ቴክ ኢንዱስትሪ ... ሕንድ ከዩናይትድ ስቴትስ (ዩ ኤስ) ቀጥሎ ... edtech ... Central Square Foundation ... Nigerian Edtech ecosystem — Indian/Nigerian edtech article — entirely irrelevant to Ethiopian clinical translation
D43	ፀረ-ተሕዋሲያን - ፀረ-ባክቴሪያ፣ ፀረ-ቫይረስ፣ ፀረ-ፈንገስ ... 'ሱፐርቦግስ(superbugs)' — AMR article with English loanword "superbugs" retained in Amharic — illustrates translationese patterns
D47	CCHF ወረርሽኞች ቫይረሱ ወደ ወረርሽኞች ሊያመራ ስለሚችል ከፍተኛ የሞት መጠን (ከ10-40%)... ህክምና፡- የክራይሚያ-ኮንጎ ... ፀረ-ቫይረስ መድሃኒት ... ribavirin — CCHF treatment article in Amharic including "ribavirin" (transliterated drug name) — medical terminology rendering pattern visible
WEB-10	JMIR 2025 scoping review identified only 3 Amharic NLP-for-public-health studies; none evaluated Ethiopia-endemic disease terminology (https://pmc.ncbi.nlm.nih.gov/articles/PMC11923465/)

Information Gaps

- Quantitative breakdown of how many of the 334 health documents are deployment-relevant (maternal-health / vaccination / clinical-guideline) vs. policy/financing/non-Ethiopian-endemic
- Whether the AFRIDOC-MT 'terminology harmonization' step (Q121) was actually completed and what reference materials were used

- Whether translationese effects in Amharic are systematic enough to bias automated metric scores against systems producing more natural Amharic

Requires Expert Verification

- MOH clinical-translator review of a sample of AFRIDOC-MT Amharic translations against MOH-distributed materials to assess register alignment
 - Comparison of WHO source-document language against actual Ethiopian MOH maternal-health booklets and clinical practice guidelines
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Remediation

Gap 1: WHO-sourced content includes substantial geographically remote material (Afghanistan polio, CCHF, volcanoes) and lacks Ethiopia-specific endemic-disease, calendar, and referral-pathway content; translationese effects are visible in Amharic translations.

Recommendation: Construct an Ethiopia-specific evaluation supplement of ~50–100 source documents drawn from actual MOH maternal-health booklets, vaccination schedules, and clinical practice guidelines — covering endemic diseases (malaria, TB, kala-azar, trachoma), Ethiopian calendar dates, and tiered-referral terminology. Use AFRIDOC-MT only for general-MT signal and weight Ethiopia-supplement scores higher in deployment decisions.

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: AFRIDOC_MT | Context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Both benchmark and deployment are text-only Amharic in Ethiopic/Ge'ez script; this is a strong form match. The benchmark explicitly handles Amharic's high token counts due to non-Latin script and increased the Amharic token limit during evaluation [Q47, Q127, Q128, Q161, Q162]. Document-level structure is preserved with NLTK + linguist sentence segmentation [Q19–Q21] and pseudo-document chunking (k=10) [Q48, Q49] to fit model context limits. Dataset inspection confirms full Ethiopic-script Amharic documents are present with multi-paragraph structure [DATASET-D1, D3]. Minor concerns: full-document translation is infeasible for most open LLMs given 8,192-token context [Q88, Q89], imposing a structural ceiling; the benchmark uses CSV with paragraph separator conventions [Q118, Q119, Q122] which may differ from MOH document formats. With IF marked LOWER priority by the user (no modality/script mismatch), these concerns do not warrant a lower score.

ID	Evidence
Q19	"While our corpus is initially structured at the article level, we aim to make it suitable for sentence-level translation tasks as well." (p.3)
Q20	"To achieve this, we segmented the raw articles into sentences using NLTK (Bird et al., 2009)." (p.3)
Q21	"To ensure high segmentation quality, we recruited a linguist and a professional translator to verify the correctness of the segmentation and make corrections as needed." (p.3)
Q47	"An initial analysis showed that some models were unable to process entire documents due to input length limits, which were exceeded by token counts in some languages (Amharic and Yorùbá)." (p.5)
Q48	"To address this, we adopted a similar approach to Lee et al. (2022), splitting documents into fixed-size chunks of k sentences to fit within token limits; the final chunk may contain fewer than k sentences." (p.5)
Q49	"To select an appropriate chunk size, we conducted initial tests with k = 1 (sentence-level), 5, 10, and 25, choosing k = 10 for our experiments." (p.5)
Q88	"Except for LLama3.1, all other open LLMs have a context length of 8192 tokens, while encoder-decoder models were primarily based on T5." (p.9)
Q89	"This makes it difficult to use the context length beyond a certain limit, making full document translation infeasible." (p.9)
Q118	"Each file contains 2500 sentences, and they are named in the format of a serial number followed by your first name." (p.18)
Q119	"Please do not delete double empty rows, as they serve to separate paragraphs. Also, avoid deleting any rows, columns, or provided text." (p.18)
Q122	"The files are in .csv format, and you can open them using Google Sheets or Microsoft Excel (for offline work)." (p.18)
Q127	"Our analysis revealed that Amharic and Yorùbá—languages with unique characteristics such as non-Latin scripts and diacritics, respectively—had the largest average token counts across the tokenizers." (p.19)
Q128	"To accommodate both languages in our experiments, we chose pseudo-documents with k=10." (p.19)
Q161	"However, for Amharic, when translating pseudo-documents with 25 sentences and full documents, there were instances exceeding the 95th percentile derived from the training statistics." (p.21)
Q162	"Therefore, we increased the token limit specifically for Amharic." (p.21)
D1	የሞቃታማ ዐውሎ ነፋሶች ... ለሞቃታማ አውሎ ነፋሶች፣ እንዲሁም ሀይለኛ አውሎ ነፋሶ ወይም አውሎ ነፋሶች ... ኃይለኛ ክብ አውሎ ነፋሶች ... 233 000 ሰዎችን ገድለዋል — Full Amharic document on tropical cyclones (WHO article), multi-sentence, Ethiopic script, with clinical/epidemiological content and specific numbers

D3	አካል ጉዳተኝነት ሰው የመሆን አካል ነው። ሁሉም ሰው ማለት ይቻላል በሕይወታቸው ላይ በሆነ ወቅት ጊዜያዊ ወይም ዘላቂ የአካል ጉዳት ያጋጥማቸዋል — Amharic WHO article on disability — prose narrative, full document preserved
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Strengths

- Amharic in Ethiopic/Ge'ez script is natively present and structurally accommodated, including increased token limits to handle Amharic's high token counts [Q127, Q128, Q162]
- Document-level structure is preserved with linguist-verified sentence segmentation [Q20, Q21]
- Pseudo-document chunking enables evaluation under realistic context-length constraints [Q48, Q49, DATASET-D1]
- Multi-parallel structure provides English↔Amharic source-target rows directly usable for the deployment direction [DATASET-D12]

ID	Evidence
Q20	"To achieve this, we segmented the raw articles into sentences using NLTK (Bird et al., 2009)." (p.3)
Q21	"To ensure high segmentation quality, we recruited a linguist and a professional translator to verify the correctness of the segmentation and make corrections as needed." (p.3)
Q48	"To address this, we adopted a similar approach to Lee et al. (2022), splitting documents into fixed-size chunks of k sentences to fit within token limits; the final chunk may contain fewer than k sentences." (p.5)
Q49	"To select an appropriate chunk size, we conducted initial tests with k = 1 (sentence-level), 5, 10, and 25, choosing k = 10 for our experiments." (p.5)
Q127	"Our analysis revealed that Amharic and Yorùbá—languages with unique characteristics such as non-Latin scripts and diacritics, respectively—had the largest average token counts across the tokenizers." (p.19)
Q128	"To accommodate both languages in our experiments, we chose pseudo-documents with k=10." (p.19)
Q162	"Therefore, we increased the token limit specifically for Amharic." (p.21)
D1	የሞቃታማ ዐውሎ ነፋሶች ... ለሞቃታማ አውሎ ነፋሶች፣ እንዲሁም ሀይለኛ አውሎ ነፋስ ወይም አውሎ ነፋሶች ... ኃይለኛ ክብ አውሎ ነፋሶች ... 233 000 ሰዎችን ገድለዋል — Full Amharic document on tropical cyclones (WHO article), multi-sentence, Ethiopic script, with clinical/epidemiological content and specific numbers
D12	"OCV ni zana ambayo hutumiwa pamoja na hatua za kudhibiti kipindupindu." (sw) / "OCV jé ohun èlò fún àfikún àwọn ilàna fún sísè àmójúótó àrùn onígá méjì" (yo) — Sentence-level multi-language row — all six languages present including Amharic

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution matches: text in Ethiopic script for Amharic. The benchmark documents that Amharic and Yorùbá produce the largest tokenizer counts and adjusted the Amharic token limit accordingly [Q127, Q161, Q162]. Dataset inspection confirms full Ethiopic-script multi-paragraph Amharic [DATASET-D1, D3, D5].

IF-2: MOH/Addis Ababa-side infrastructure can support Unicode Ethiopic-script processing; deployment is text-only with print and digital distribution [deployment context]. End-user connectivity (19.4% internet penetration nationally [WEB-11]) is relevant for distribution but not for the MT system's input form.

IF-3: Document-level format with paragraph preservation is supported via CSV with double-empty-row paragraph separators [Q118, Q119, Q122]. However, MOH distribution materials may use formats (Word, PDF, print-ready layouts) requiring additional conversion. The benchmark's full-document translation is infeasible for most open LLMs (8,192-token ceiling) [Q88,

Q89], constraining how the system handles longer MOH documents — a real but moderate constraint.

IF-4: Minor form mismatches: (1) full-document translation infeasibility for most LLMs forces pseudo-document chunking, which the benchmark itself shows can degrade African-language translation quality vs. sentence-level [Q61, Q72]; (2) CSV-with-paragraph-separator input format differs from typical MOH document formats but is a reasonable proxy for document-level structure.

ID	Evidence
Q61	“Pseudo-document translation is worse than sentence-level translation when translating into African languages Our results from pseudo-document translation show a performance drop across different models compared to sentence-level translation, especially when translating into African languages.” (p.6)
Q72	“Human evaluation results align closely with d-chrF, which favors sentence-level translations over pseudo-document translations when translating into African languages.” (p.8)
Q88	“Except for LLama3.1, all other open LLMs have a context length of 8192 tokens, while encoder-decoder models were primarily based on T5.” (p.9)
Q89	“This makes it difficult to use the context length beyond a certain limit, making full document translation infeasible.” (p.9)
Q118	“Each file contains 2500 sentences, and they are named in the format of a serial number followed by your first name.” (p.18)
Q119	“Please do not delete double empty rows, as they serve to separate paragraphs. Also, avoid deleting any rows, columns, or provided text.” (p.18)
Q122	“The files are in .csv format, and you can open them using Google Sheets or Microsoft Excel (for offline work).” (p.18)
Q127	“Our analysis revealed that Amharic and Yorùbá—languages with unique characteristics such as non-Latin scripts and diacritics, respectively—had the largest average token counts across the tokenizers.” (p.19)
Q161	“However, for Amharic, when translating pseudo-documents with 25 sentences and full documents, there were instances exceeding the 95th percentile derived from the training statistics.” (p.21)
Q162	“Therefore, we increased the token limit specifically for Amharic.” (p.21)
D1	የሞቃታማ ዐውሎ ነፋሶች ... ለሞቃታማ አውሎ ነፋሶች፣ እንዲሁም ሀይለኛ አውሎ ነፋስ ወይም አውሎ ነፋሶች ... ኃይለኛ ክብ አውሎ ነፋሶች ... 233 000 ሰዎችን ገድለዋል — Full Amharic document on tropical cyclones (WHO article), multi-sentence, Ethiopic script, with clinical/epidemiological content and specific numbers
D3	አካል ጉዳተኝነት ሰው የመሆን አካል ነው። ሁሉም ሰው ማለት ይቻላል በሕይወታቸው ላይ በሆነ ወቅት ጊዜያዊ ወይም ዘላቂ የአካል ጉዳት ያጋጥማቸዋል — Amharic WHO article on disability — prose narrative, full document preserved
D5	ፀረ-ተሕዋስያንን መቋቋም(AMR) ... ባክቴሪያ፣ ጥገኛ ተውሳኮች፣ ቫይረሶችና በፈንገሶች ምክንያት — Amharic AMR article — specialized medical terminology in Ethiopic script
WEB-11	Ethiopia internet penetration ~19.4% in early 2024 — affects distribution channel but not MT system input form (https://datareportal.com/reports/digital-2024-ethiopia)

Information Gaps

- Whether actual MOH source documents fit within the benchmark’s pseudo-document (k=10) chunking assumption or routinely exceed it

Requires Expert Verification

- Confirmation that MOH document formats (Word/PDF/print) can be losslessly converted to the benchmark’s text-with-paragraph-separator representation

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: AFRIDOC_MT | Context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's output ontology operationalizes translation quality via d-BLEU, d-chrF, GPT-4o-as-judge with four explicit quality dimensions (fluency, content errors, lexical cohesion errors, grammatical cohesion errors) [Q179, Q180, Q183, Q184, Q189, Q190], and human DA on 0–100 scale [Q195, Q196]. The deployment's primary success criterion — MOH/EFDA glossary compliance — is not operationalized by any benchmark scoring function: d-BLEU/d-chrF and human DA reward similarity to the AFRIDOC-MT references and are agnostic to institutional terminology lists, and the planned terminology harmonization step (Q121) is not documented as completed. Document-level terminological consistency is partially captured by the lexical cohesion (LE) error category in GPT-4o-as-judge [Q189], but GPT-4o-as-judge is documented as unreliable for Amharic [Q99, Q218, Q219]. Faithfulness (CE) and readability (fluency) are reasonably covered. With OO marked HIGH priority by the user and glossary compliance the deployment's primary criterion, the score is 2: the benchmark's decision rules systematically misrepresent the deployment's central success criterion.

ID	Evidence
Q99	"GPT-4o was employed to assess and rate the translation outputs of four models. While its ratings were consistent for translations into English, the same was not observed for translations into African languages, likely due to the model's limited understanding of these languages." (p.10)
Q121	"We will provide a list of extracted terminologies soon so that you can harmonize how terminologies are translated." (p.18)
Q179	"We compared translations performed at the sentence-level and pseudo-document level in terms of fluency, content errors, and cohesion errors—specifically lexical (LE) and grammatical (GE) errors—using the same definitions as Sun et al. (2025)." (p.22)
Q180	"GPT-4o is prompted to rate the fluency of a document on a scale from 1 to 5, where 5 indicates high fluency and 1 represents low fluency." (p.22)
Q183	"GPT-4 is prompted to identify and list the mistakes, such as incorrect translations, omissions, additions, and any other errors, by comparing the model's output to the reference translation." (p.22)
Q184	"After identifying these errors, we count all of them and compute the average across all documents, reporting that as the content error (CE)." (p.22)
Q189	"Lexical Cohesion Mistakes: Issues with vocabulary usage, incorrect or missing synonyms, or overuse of certain words that disrupt the flow." (p.23)
Q190	"Grammatical Cohesion Mistakes: Problems with pronouns, conjunctions, or grammatical structures that link sentences and clauses." (p.23)
Q195	"Each annotator was assigned 80 documents to evaluate, tasked with marking as many error spans as possible and rating the overall quality on a scale from 0 to 100." (p.24)
Q196	"This annotation followed the error span annotation (ESA) (Kocmi et al., 2024) protocol as implemented within the Appraise Evaluation Framework (Federmann, 2018)." (p.24)
Q218	"These findings are consistent with the d-chrF scores for the models, but they do not align with the evaluations from GPT-4o as a judge." (p.26)
Q219	"These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages." (p.26)

Strengths

- Faithfulness to source meaning is well operationalized via content errors (CE) in GPT-4o-as-judge and human DA [Q183, Q184]
- Cohesion errors taxonomy (lexical and grammatical) explicitly recognizes document-level discourse phenomena [Q187, Q188, Q189, Q190]
- chrF prioritized over BLEU because it captures Amharic morphological richness — a sound choice for the target language [Q53]
- Multiple complementary scoring streams (lexical, neural, GPT-4o judge, human DA, qualitative) provide breadth [Q83, Q98]

ID	Evidence
Q53	“We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C.” (p.5)
Q83	“Outputs were evaluated using standard MT metrics, GPT-4o as a judge, and human direct assessment.” (p.9)
Q98	“However, the scores were nearly identical across all models, offering no meaningful differentiation. Hence, for our document-level evaluation, in addition to lexical-based metrics, we incorporated three other evaluation approaches: using GPT-4o as a judge, human evaluation, and qualitative analysis.” (p.10)
Q183	“GPT-4 is prompted to identify and list the mistakes, such as incorrect translations, omissions, additions, and any other errors, by comparing the model’s output to the reference translation.” (p.22)
Q184	“After identifying these errors, we count all of them and compute the average across all documents, reporting that as the content error (CE).” (p.22)
Q187	“Cohesion refers to how different parts of a text are connected using language structures like grammar and vocabulary.” (p.23)
Q188	“It ensures that sentences flow smoothly and the text makes sense as a whole.” (p.23)
Q189	“Lexical Cohesion Mistakes: Issues with vocabulary usage, incorrect or missing synonyms, or overuse of certain words that disrupt the flow.” (p.23)
Q190	“Grammatical Cohesion Mistakes: Problems with pronouns, conjunctions, or grammatical structures that link sentences and clauses.” (p.23)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output categories are continuous/count metrics (d-BLEU, d-chrF, fluency 1–5, CE/LE/GE counts, DA 0–100) [Q180, Q191]. Regional relevance is partial: faithfulness and fluency are universally relevant; cohesion errors capture some terminological consistency; glossary compliance is absent.

OO-2: Missing categories specific to the deployment: (a) MOH/EFDA glossary-compliance score — primary deployment success criterion is unoperationalized; (b) document-level term-consistency score that explicitly checks whether the same source term is rendered identically (LE captures vocabulary errors but not term-by-term institutional consistency); (c) readability calibrated for semi-literate patients — fluency 1–5 is not population-calibrated [DATASET CRITICAL Concern 2]; (d) Ethiopia-specific terminology coverage (endemic diseases, calendar, referral pathways).

OO-3: The cohesion-error dimensions are operationalized via GPT-4o prompts that compare model output to reference translation [Q185, Q186] — implicitly encoding the AFRIDOC-MT translators’ choices as the standard, which may not align with MOH/EFDA approved terminology [DATASET-D5, D43, D47]. Fluency rated by GPT-4o without a reference [Q181]

embeds whatever Amharic register GPT-4o has internalized, which may differ from Ethiopian clinical Amharic.

OO-4: Stakeholder-driven taxonomy redesign would be high value: adding (a) glossary-conformance scoring against MOH/EFDA term lists, (b) explicit term-consistency tracking, (c) MOH-translator-judged faithfulness, and (d) readability for semi-literate audiences would address the most consequential OO gaps. The Academy of Ethiopian Languages Amharic Medical Dictionary [WEB-8] and a multilingual glossary [WEB-9] exist but are not integrated.

OO-5: Structural validity is moderately compromised because the central deployment construct (institutional terminology compliance) is not represented in the scoring functions [DATASET CRITICAL Concern 2]. Content validity is also harmed by missing categories. External validity is at risk: high benchmark scores would not necessarily predict MOH-acceptable outputs.

ID	Evidence
Q121	"We will provide a list of extracted terminologies soon so that you can harmonize how terminologies are translated." (p.18)
Q180	"GPT-4o is prompted to rate the fluency of a document on a scale from 1 to 5, where 5 indicates high fluency and 1 represents low fluency." (p.22)
Q181	"This evaluation is conducted without providing any reference document." (p.22)
Q183	"GPT-4 is prompted to identify and list the mistakes, such as incorrect translations, omissions, additions, and any other errors, by comparing the model's output to the reference translation." (p.22)
Q185	"Cohesion: GPT-4 is prompted to rate cohesion-related mistakes, including lexical and grammatical errors, in the model's output, comparing it to the reference translation." (p.23)
Q186	"We count each error individually, compute the average across the documents, and report them as lexical errors (LE) and grammatical errors (GE)." (p.23)
Q189	"Lexical Cohesion Mistakes: Issues with vocabulary usage, incorrect or missing synonyms, or overuse of certain words that disrupt the flow." (p.23)
Q190	"Grammatical Cohesion Mistakes: Problems with pronouns, conjunctions, or grammatical structures that link sentences and clauses." (p.23)
Q191	"Fluency can only have values between 1 and 5. However, the other metrics, including CE, GE, and LE, do not have a specific range and can take on any value because they are counts." (p.24)
D5	ፀረ-ተሕዋስንን መቋቋም(AMR) ... ባክቴሪያ፣ ጥገኛ ተውሳኮች፣ ቫይረሶችና በፈንገሶች ምክንያት — Amharic AMR article — specialized medical terminology in Ethiopic script
D43	ፀረ-ተሕዋሲያን - ፀረ-ባክቴሪያ፣ ፀረ-ቫይረስ፣ ፀረ-ፈንገስ ... 'ሱፐርባግስ(superbugs)' — AMR article with English loanword "superbugs" retained in Amharic — illustrates translationese patterns
D47	CCHF ወረርሽኞች ቫይረሱ ወደ ወረርሽኞች ሊያመራ ስለሚችል ከፍተኛ የሞት መጠን (ከ10-40%)... ህክምና፡- የክራይሚያ-ኮንጎ ... ፀረ-ቫይረስ መድሃኒት ... ribavirin — CCHF treatment article in Amharic including "ribavirin" (transliterated drug name) — medical terminology rendering pattern visible
WEB-8	Academy of Ethiopian Languages Amharic Medical Dictionary exists as a potentially relevant terminology reference for re-annotation (https://archive.org/details/amharic-dictionary-of-medical-terms-108p-copy)
WEB-9	Multilingual English-Amharic-Oromiffa-Somali-Afar health glossary exists on Scribd but is not integrated (https://www.scribd.com/document/416080918/English-Amharic-Oromiffa-Somali-Afar)

Information Gaps

- Whether the AFRIDOC-MT planned terminology harmonization step [Q121] was completed and what reference list was used
- Whether MOH-distributed terminology in Amharic clinical guidelines is actually formalized as a single canonical list, or whether multiple competing references exist

Requires Expert Verification

- MOH/EFDA confirmation of which Amharic glossary is the authoritative compliance reference
- Whether AFRIDOC-MT Amharic translators followed any institutional glossary during translation

Remediation

Gap 1: No benchmark scoring function operationalizes MOH/EFDA glossary compliance — the deployment's primary success criterion is unmeasured.

Recommendation: Build a glossary-conformance scorer that extracts source clinical terms, looks them up in the Academy of Ethiopian Languages Amharic Medical Dictionary [WEB-8] and any EFDA-approved list, and scores system outputs on percentage of terms rendered with the institutionally approved Amharic equivalent. Track this metric alongside d-chrF on the Ethiopian-specific evaluation set.

Gap 2: Readability for semi-literate patients is not calibrated by the benchmark — fluency 1–5 reflects expert-translator perception, not patient comprehension.

Recommendation: Add a readability evaluation track using either an Amharic readability formula adaptation or HEW-mediated comprehension testing on a small patient sample for maternal-health booklet outputs. Report this as a separate metric alongside fluency scores.

ID	Evidence
WEB-8	Academy of Ethiopian Languages Amharic Medical Dictionary exists as a potentially relevant terminology reference for re-annotation (https://archive.org/details/amharic-dictionary-of-medical-terms-108p-copy)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: AFRIDOC_MT | Context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output content (the reference translations and human evaluation labels themselves) raises the most consequential validity concerns. The four named Amharic translators [Q109] are recruited via a language coordinator [Q23, Q24] and attended a domain workshop with short translation guidelines [Q26], but the benchmark documents no MOH credentials, clinical backgrounds, EFDA training, or institutional affiliations — the deployment explicitly designates MOH-trained clinical translators as primary ground-truth authority and Amharic-speaking physicians as fallback. Web search did not find public credentials for these translators or a public MOH clinical translation certification registry [WEB-3 register variation; benchmark_validity_gaps_summary annotator_credentials NOT FOUND]. Inter-annotator agreement for Amharic human DA is $\alpha=0.46$ [Q211, Q71], the lowest of the five languages, indicating noisy reference quality signal even within the benchmark's own framework. The translators' regional affiliation is undocumented, leaving regional Amharic register variation across Tigray/Oromia/Somali bureaus unaddressed. With OC marked HIGH priority and the user explicitly excluding general translators and academic linguists from authoritative status, the alignment is poor — score 2.

ID	Evidence
Q23	"We translated the extracted 10k English sentences to the 5 African languages through 4 expert translators per language." (p.3)
Q24	"The translators were recruited through a language coordinator who is also a native speaker of the language." (p.3)
Q26	"Due to the domain-specific nature of the task, before starting the translation process, we conducted a translation workshop, during which three translation experts shared their experiences in creating terminologies and they also shared existing resources with the translators including short translation guidelines (Appendix A.1)." (p.3)
Q71	"We obtained Krippendorff's alpha values of ≥ 0.40 , which are relatively low due to the fine granularity of the evaluation scale." (p.8)
Q109	"Also, we also appreciate the translators whose names we have listed below: 1. Amharic: Bereket Tilahun, Hana M. Tamiru, Biniam Asmlash, Lidetewold Kebede 2. Hausa: Junaid Garba, Umma Abubakar, Jibrin Adamu, Ruqayya Nasir Iro 3. Swahili: Mohamed Mwinyimkuu, Laanyu koone, Baraka Karuli Mgasa, Said Athumani Said 4. Yorùbá: Ifeoluwa Akinsola, Simeon Onaolapo, Ganiyat Dasola Afolabi, Oluwatosin Koya 5. Zulu: Busisiwe Pakade, Rooweither Mabuya, Tholakele Zungu, Zanele Thembani" (p.11)
Q211	"We obtained α scores of 0.46, 0.57, 0.40, and 0.81, and 0.54 for Amharic, Hausa, Swahili, Yorùbá, and Zulu respectively." (p.26)
WEB-3	Amharic dialects mutually intelligible per Wikipedia, but health-register variation across bureaus is a separate question — unresolved (https://en.wikipedia.org/wiki/Amharic)

Strengths

- Four expert translators per language with a workshop on domain-specific terminology and short translation guidelines [Q23, Q26]
- Use of MT engines was prohibited and quality assurance described as capable of detection [Q120]
- AfriCOMET-based QE with manual review threshold (0.65) added a layer of automated quality control [Q27, Q28, Q29]
- Translators paid at clearly disclosed rates (\$1,250 per 2,500 sentences, \$55.15 per 80 documents for evaluators) — transparent compensation [Q31, Q198]
- Dataset analysis confirms specialized clinical terminology is rendered in Ethiopic script across complex topics [DATASET-D5, D43, D47]

ID	Evidence
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Q23	"We translated the extracted 10k English sentences to the 5 African languages through 4 expert translators per language." (p.3)
Q26	"Due to the domain-specific nature of the task, before starting the translation process, we conducted a translation workshop, during which three translation experts shared their experiences in creating terminologies and they also shared existing resources with the translators including short translation guidelines (Appendix A.1)." (p.3)
Q27	"Quality control was conducted using automated quality estimation, followed by manual inspections by our language coordinators." (p.3)
Q28	"We also used Quality Estimation (QE), specifically AfriCOMET (Wang et al., 2024a), Given a translated sentence in any African language and its corresponding source English sentence, AfriCOMET generates a score between 0 and 1, where 0 indicates poor quality and higher values signify better quality." (p.3)
Q29	"The translators, in collaboration with the language coordinators, were tasked with reviewing instances that had quality estimation scores below 0.65." (p.3)
Q31	"Each translator was paid \$1, 250 for 2, 500 sentences." (p.3)
Q120	"Use the language field to input the translations. It is essential not to rely on translation engines, as our quality assurance process can detect this. Depending on such tools may result in potential issues that you would need to address, leading to additional work on your part." (p.18)
Q198	"Each annotator was remunerated with \$55.15" (p.24)
D5	ፀረ-ተሕዋስያንን መቋቋም(AMR) ... ባክቴሪያ፣ ጥገኛ ተውሳኮች፣ ቫይረሶችና በፈንገሶች ምክንያት — Amharic AMR article — specialized medical terminology in Ethiopic script
D43	ፀረ-ተሕዋሲያን - ፀረ-ባክቴሪያ፣ ፀረ-ቫይረስ፣ ፀረ-ፈንገስ ... 'ሱፐርባግስ(superbugs)' — AMR article with English loanword "superbugs" retained in Amharic — illustrates translationese patterns
D47	CCHF ወረርሽኞች ቫይረሱ ወደ ወረርሽኞች ሊያመራ ስለሚችል ከፍተኛ የሞት መጠን (ከ10-40%)... ህክምና፡- የክራይሚያ-ኮንጎ ... ፀረ-ቫይረስ መድሃኒት ... ribavirin — CCHF treatment article in Amharic including "ribavirin" (transliterated drug name) — medical terminology rendering pattern visible

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

- OC-1:** Ground-truth labels (reference translations and human DA scores) likely do NOT reflect the deployment's authoritative regional stakeholder population. The user designates MOH-trained clinical translators as primary and Amharic-speaking physicians as fallback ground-truth; the benchmark documents neither MOH credentials nor clinical/medical backgrounds for the four named Amharic translators [Q109]. No regional bureau affiliation is documented [register_variation web search NOT FOUND].
- OC-2:** Substantial disagreement potential between AFRIDOC-MT translators and MOH-credentialed clinical translators is plausible: the benchmark's translators may have used different terminology choices for AMR, drug names, infant feeding terms than MOH/EFDA standards prescribe [DATASET-D5, D43, D47, D44]. The $\alpha=0.46$ Amharic agreement [Q211] further indicates that even the benchmark's own annotation pool produced noisy quality judgments — the lowest among five languages.
- OC-3:** Annotator demographics in the benchmark are minimally documented: names of four translators per language [Q109], recruited through a 'language coordinator who is also a native speaker' [Q24]. No professional credentials, institutional affiliations, regions, ages, or clinical backgrounds are disclosed. No formal datasheet/data statement detailing demographics is

provided. This is a clear documentation gap relative to standard practice.

OC-4: Re-annotation by an MOH-credentialed clinical translator pool would be high-value: producing parallel reference translations or post-editing AFRIDOC-MT references to MOH/EFDA standards would yield a deployment-aligned reference set. Rough cost estimate scales from disclosed per-sentence rates [Q31].

OC-5: Aggregation methods: the benchmark uses single reference translations per row (one of four translators), and human DA averages three annotators per language [Q69, Q194]. Krippendorff's alpha is reported but no explicit handling of minority-perspective preservation is described. Given $\alpha=0.46$ for Amharic, averaging may erase substantial annotator-level disagreement.

OC-6: Convergent validity is at risk: AFRIDOC-MT references may not converge with MOH-credentialed translator judgments [DATASET CRITICAL Concern 1]. External validity is at risk: high system scores against AFRIDOC-MT references do not reliably predict MOH-acceptable performance. Construct validity for the regional deployment is materially weakened by both annotator-credentials uncertainty and low Amharic IAA [Q211, DATASET CRITICAL Concern 4].

ID	Evidence
Q23	"We translated the extracted 10k English sentences to the 5 African languages through 4 expert translators per language." (p.3)
Q24	"The translators were recruited through a language coordinator who is also a native speaker of the language." (p.3)
Q26	"Due to the domain-specific nature of the task, before starting the translation process, we conducted a translation workshop, during which three translation experts shared their experiences in creating terminologies and they also shared existing resources with the translators including short translation guidelines (Appendix A.1)." (p.3)
Q31	"Each translator was paid \$1, 250 for 2, 500 sentences." (p.3)
Q69	"In Table 10 we report average direct assessment (DA) scores (on a scale from 0 to 100) from three annotators per language for the health domain, when translating into four African languages." (p.7)
Q109	"Also, we also appreciate the translators whose names we have listed below: 1. Amharic: Bereket Tilahun, Hana M. Tamiru, Biniam Asmlash, Lidetewold Kebede 2. Hausa: Junaid Garba, Umma Abubakar, Jibrin Adamu, Ruqayya Nasir Iro 3. Swahili: Mohamed Mwinyimkuu, Laanyu koone, Baraka Karuli Mgasa, Said Athumani Said 4. Yorùbá: Ifeoluwa Akinsola, Simeon Onaolapo, Ganiyat Dasola Afolabi, Oluwatosin Koya 5. Zulu: Busisiwe Pakade, Rooweither Mabuya, Tholakele Zungu, Zanele Them bani" (p.11)
Q194	"For the human evaluation, three native speakers of the African languages—primarily translators involved in the dataset creation—were recruited." (p.24)
Q211	"We obtained α scores of 0.46, 0.57, 0.40, and 0.81, and 0.54 for Amharic, Hausa, Swahili, Yorùbá, and Zulu respectively." (p.26)
D5	ፀረ-ተሕዋስንን ወደቋም(AMR) ... ባክቴሪያ፣ ጥገኛ ተውሳኮች፣ ቫይረሶችና በፈንገሶች ምክንያት — Amharic AMR article — specialized medical terminology in Ethiopic script
D43	ፀረ-ተሕዋሲያን - ፀረ-ባክቴሪያ፣ ፀረ-ቫይረስ፣ ፀረ-ፈንገስ ... 'ሱፐርቦግስ(superbugs)' — AMR article with English loanword "superbugs" retained in Amharic — illustrates translationese patterns
D44	ጨቅላ ህፃናት ከ6 ወር ጀምሮ የተጣራ፣ የተፈጨ እና ከፊል ጠጣር ምግቦችን መመገብ ይችላሉ — Amharic instruction for pureed/mashed foods from 6 months — clinically precise maternal-health instruction
D47	CCHF ወረርሽኞች ቫይረሱ ወደ ወረርሽኞች ሊያመራ ስለሚችል ከፍተኛ የሞት መጠን (ከ10-40%)... ህክምና፡- የክራይሚያ-ኮንጎ ... ፀረ-ቫይረስ መድሃኒት ... ribavirin — CCHF treatment article in Amharic including "ribavirin" (transliterated drug name) — medical terminology rendering pattern visible
WEB-8	Academy of Ethiopian Languages Amharic Medical Dictionary exists as a potentially relevant terminology reference for re-annotation (https://archive.org/details/amharic-dictionary-of-medical-terms-108p-copy)

Information Gaps

- Professional credentials, MOH affiliation, clinical training, and regional location of the four named Amharic translators
- Whether the human DA evaluators (three native speakers per language, primarily drawn from the translation team [Q194]) hold any clinical credentials
- Whether the planned terminology harmonization [Q121] used any MOH/EFDA-aligned source
- Why Amharic IAA ($\alpha=0.46$) is so much lower than Yorùbá (0.81) — could reflect translation ambiguity, register variation, or fine granularity of the scale

Requires Expert Verification

- Direct inquiry to AFRIDOC-MT authors (e.g., Israel Abebe Azime, Saarland University) regarding Amharic translator credentials and regional affiliations
 - MOH clinical-translator side-by-side review of a sample of AFRIDOC-MT Amharic references to assess alignment with MOH standards
 - Stakeholder elicitation from regional health bureau staff (Tigray, Oromia, Somali) on whether AFRIDOC-MT Amharic register is acceptable
-

Remediation

Gap 1: AFRIDOC-MT Amharic translators have no documented MOH credentials or clinical backgrounds, and $\alpha=0.46$ Amharic IAA indicates noisy reference signal — a direct mismatch with the deployment's MOH-trained-translator ground-truth authority.

Recommendation: Recruit 2–3 MOH-credentialed clinical translators (or Amharic-speaking physicians as fallback) to post-edit a sampled subset (e.g., 50–100 documents) of AFRIDOC-MT Amharic references to MOH/EFDA standards, producing a deployment-aligned evaluation set. Document each post-editor's MOH affiliation, clinical credentials, and regional bureau experience.

Gap 2: Regional Amharic register variation across Tigray, Oromia, Amhara, and Somali bureaus is unaddressed; benchmark uses single Addis Ababa-centric references whose regional acceptance is unverified.

Recommendation: Conduct stakeholder elicitation with regional health bureau staff in at least Oromia, Tigray, and Somali regions to identify any region-specific terminology preferences that would diverge from Addis Ababa-centric references. Document any findings as a register-variation flag set for evaluation.

Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: AFRIDOC_MT | Context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is text-only Amharic translation, matching the deployment exactly [Q1, Q44]. The benchmark provides comprehensive output-form scoring infrastructure: d-BLEU and d-chrF with bootstrap CI [Q51, Q52], chrF prioritized for African morphological richness [Q53], full results tables across models/domains/granularities [Q226–Q231], and per-prompt sensitivity analyses [Q236–Q239]. AfriCOMET is excluded from document-level scoring due to its 512-token context limit [Q175, Q176], and the long-context QE model (ModernBERT) showed no meaningful differentiation and lacks Amharic training [Q97, Q98]. GPT-4o-as-judge is explicitly documented as inconsistent for African languages including Amharic [Q99, Q218, Q219], independently corroborated [WEB-17, WEB-18]. With OF marked LOWER priority by the user (modality match), score 4: form is well-matched, but document-level neural QE for Amharic remains an open gap.

ID	Evidence
Q1	"This paper introduces AFRIDOC-MT, a document-level multi-parallel translation dataset covering English and five African languages: Amharic, Hausa, Swahili, Yorùbá, and Zulu." (p.1)
Q44	"Given that our created dataset can be used for sentence-level translation and as a baseline for document-level translation, we evaluate all models on the test splits for each domain." (p.5)
Q51	"Hence, we realigned sentence-level or pseudo-translation outputs into full documents, then computed BLEU (Papineni et al., 2002) and chrF (Popović, 2015) to create document BLEU (d-BLEU) and document chrF (d-chrF)." (p.5)
Q52	"Metrics were computed using SacreBLEU (Post, 2018) with bootstrap resampling (n = 1000) to report 95% confidence intervals." (p.5)
Q53	"We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C." (p.5)
Q97	"However, we evaluated the document-level translation outputs using ModernBERT-base-long-context-qe-v1, trained on the WMT human evaluation dataset across 41 language pairs, including over 20 languages and three African languages (Hausa, Xhosa, and Zulu), two of which are included to our work." (p.10)
Q98	"However, the scores were nearly identical across all models, offering no meaningful differentiation. Hence, for our document-level evaluation, in addition to lexical-based metrics, we incorporated three other evaluation approaches: using GPT-4o as a judge, human evaluation, and qualitative analysis." (p.10)
Q99	"GPT-4o was employed to assess and rate the translation outputs of four models. While its ratings were consistent for translations into English, the same was not observed for translations into African languages, likely due to the model's limited understanding of these languages." (p.10)
Q175	"For evaluation, we used BLEU and chrF scores but excluded AfriCOMET due to its backbone model, AfroXLM-R-L (Alabi et al., 2022; Adelani et al., 2024a), having a context length of 512 tokens." (p.22)
Q176	"This made it impractical to compute COMET scores for document-level outputs." (p.22)
Q218	"These findings are consistent with the d-chrF scores for the models, but they do not align with the evaluations from GPT-4o as a judge." (p.26)
Q219	"These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages." (p.26)
Q226	"Table 15: Performance results of various models on the sentence-level task for the Health domain, measured using document level metric d-BLEU and d-chrF." (p.29)
Q227	"Table 16: Performance results of various models on the sentence-level task for the Tech domain, measured using document level metric d-BLEU and d-chrF." (p.29)

Q228	"Table 17: Performance results of various models on the sentence-level task for the Health domain, measured using sentence level metric s-BLEU, s-CHRF, and s-COMET." (p.30)
Q229	"Performance results of various models on the sentence-level task for the Tech domain, measured using sentence level metric s-BLEU, s-CHRF, and s-COMET." (p.31)
Q230	"Table 19: Performance results of various models on the pseudo-document-level task for the Health domain, measured using document level metric d-BLEU and d-CHRF." (p.32)
Q231	"Table 20: Performance results of various models on the pseudo-document-level task for the Tech domain, measured using document level metric d-BLEU and d-CHRF." (p.32)
Q236	"Figure 19: d-chrF scores for some LLMs for sentence-level translation using different prompts when translating into African languages" (p.37)
Q237	"Figure 20: d-chrF scores for some LLMs for sentence-level translation using different prompts when translating into African languages" (p.37)
Q238	"Figure 21: d-chrF scores for some LLMs for sentence-level translation using different prompts when translating into English" (p.37)
Q239	"Figure 22: d-chrF scores for some LLMs for sentence-level translation using different prompts when translating into English" (p.37)
WEB-17	Walia-LLM corroborates GPT-4o-as-judge unreliability for Amharic (https://aclanthology.org/2024.findings-emnlp.25/)
WEB-18	arXiv 2402.08015 confirms GPT-4 'lacks ability to evaluate most generation tasks' in Amharic (https://arxiv.org/abs/2402.08015)

Strengths

- Document-level scoring (d-BLEU, d-chrF) with bootstrap-resampling 95% CIs [Q51, Q52]
- chrF prioritized for African morphological richness — sound metric choice for Amharic [Q53]
- Multiple complementary scoring streams (lexical, neural QE, GPT-4o judge, human DA, qualitative analysis) [Q98]
- Honest reporting of GPT-4o-as-judge unreliability for African languages — directs users to human DA where reliable [Q67, Q68, Q99]
- Output text-only format matches deployment's print/digital Amharic distribution requirement

ID	Evidence
Q51	"Hence, we realigned sentence-level or pseudo-translation outputs into full documents, then computed BLEU (Papineni et al., 2002) and chrF (Popović, 2015) to create document BLEU (d-BLEU) and document chrF (d-chrF)." (p.5)
Q52	"Metrics were computed using SacreBLEU (Post, 2018) with bootstrap resampling (n = 1000) to report 95% confidence intervals." (p.5)
Q53	"We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C." (p.5)
Q67	"These inconsistencies raise concerns about GPT-4o's reliability." (p.7)
Q68	"Consequently, we focus on human evaluation going forward." (p.7)
Q98	"However, the scores were nearly identical across all models, offering no meaningful differentiation. Hence, for our document-level evaluation, in addition to lexical-based metrics, we incorporated three other evaluation approaches: using GPT-4o as a judge, human evaluation, and qualitative analysis." (p.10)
Q99	"GPT-4o was employed to assess and rate the translation outputs of four models. While its ratings were consistent for translations into English, the same was not observed for translations into African languages, likely due to the model's limited understanding of these languages." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text-only Amharic) matches deployment exactly. The deployment produces Amharic documents for print and digital distribution; benchmark evaluates text-only translation [Q1, Q44, infrastructure_notes].

OF-2: Text-to-speech is not in scope for this deployment — outputs are text-only. INSUFFICIENT DOCUMENTATION on TTS, but this is not a deployment requirement.

OF-3: Literacy considerations affect end-user readability. Ethiopia adult literacy ~51.8% (2017 UNESCO) [WEB-4] with significant urban-rural and gender gaps [WEB-6, WEB-7]. The benchmark's fluency dimension (1–5) and human DA (0–100) are not specifically calibrated for semi-literate populations — output text may be fluent at expert-translator level but inaccessible to lower-literacy patients. Partial mismatch in scoring calibration relative to deployment audience.

OF-4: External validity for output form is mostly preserved (text-to-text match). Two minor harms: (a) no validated document-level neural QE for Amharic [Q97, Q98, WEB-18], leaving d-chrF and human DA as the only scalable signals; (b) fluency calibration not tied to semi-literate patient comprehension [literacy_profile note].

ID	Evidence
Q1	"This paper introduces AFRIDOC-MT, a document-level multi-parallel translation dataset covering English and five African languages: Amharic, Hausa, Swahili, Yorùbá, and Zulu." (p.1)
Q44	"Given that our created dataset can be used for sentence-level translation and as a baseline for document-level translation, we evaluate all models on the test splits for each domain." (p.5)
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Q98	"However, the scores were nearly identical across all models, offering no meaningful differentiation. Hence, for our document-level evaluation, in addition to lexical-based metrics, we incorporated three other evaluation approaches: using GPT-4o as a judge, human evaluation, and qualitative analysis." (p.10)
WEB-4	Ethiopia adult literacy ~51.8% (2017 UNESCO) — relevant to readability calibration (https://www.theglobaleconomy.com/Ethiopia/Literacy_rate/)
WEB-6	Significant urban-rural and gender literacy gaps in Ethiopia (https://zoetalentsolutions.com/education-statistics-for-ethiopia/)
WEB-7	https://countrymeters.info/en/Ethiopia
WEB-18	arXiv 2402.08015 confirms GPT-4 'lacks ability to evaluate most generation tasks' in Amharic (https://arxiv.org/abs/2402.08015)

Information Gaps

- Whether deployment requires any TTS or audio output for low-literacy patient channels — unresolved in elicitation
- Whether MOH would require specific document formatting (PDF, Word, print-ready) beyond text

Requires Expert Verification

- Whether AFRIDOC-MT-style fluency scoring correlates with semi-literate-patient comprehension (could be tested with a small readability study)

Remediation

Gap 1: No validated document-level neural quality estimation metric exists for Amharic; AfriCOMET is sentence-level only and the long-context QE model excludes Amharic.

Recommendation: Adopt sentence-level AfriCOMET as a routine quality-control signal at translation-unit level, paired with d-chrF for document-level signal. Treat document-level neural QE as an open research question and avoid relying on GPT-4o-as-judge for Amharic outputs given the benchmark's own and independent corroborating evidence of unreliability [Q219, WEB-18].

ID	Evidence
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Q219	"These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages." (p.26)
WEB-18	arXiv 2402.08015 confirms GPT-4 'lacks ability to evaluate most generation tasks' in Amharic (https://arxiv.org/abs/2402.08015)